



United International University
QUEST FOR EXCELLENCE



CSE 1326: Digital Logic Design Lab

Ripple counter

United International University

Objective

- Implement
 - Ripple Counters: 4-bit Upward Counter using
 1. D flip-flops (IC 7474), and
 2. J-K flip-flops (IC 7476),

Counter Design

- Varies with
 - # of bits
 - Upward/Downward
 - Type of flip-flops (How to toggle)
 - +ve/-ve edge of FFs

Ripple Counter: 4-bit Upward Counter

Upward Counting Sequence

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

- Varies with
 - # of bits
 - Upward/Downward
 - Type of flip-flops (How to toggle)
 - +ve/-ve edge of FFs
- A 4-bit counter counts from 0 to 15 to 0 again and repeats.
- Need 4 FFs
- FF toggles between 0 and 1
- Need to trigger them to toggle at the right time
- Start from LSB, as that toggles most frequently. It toggles in every clock pulse.
- Hence, we directly connect Q_0 to clock.

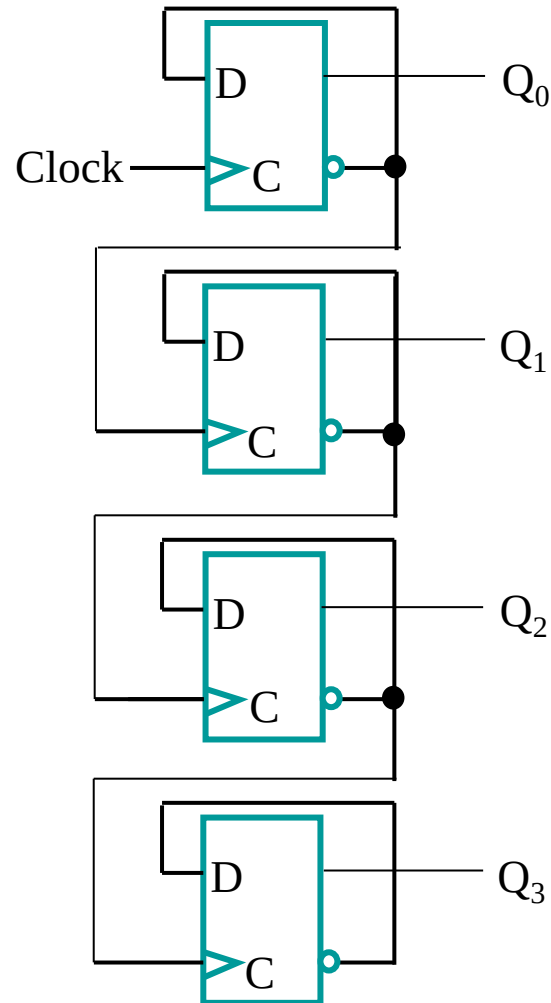
Ripple Counter: 4-bit Upward Counter

Upward Counting Sequence

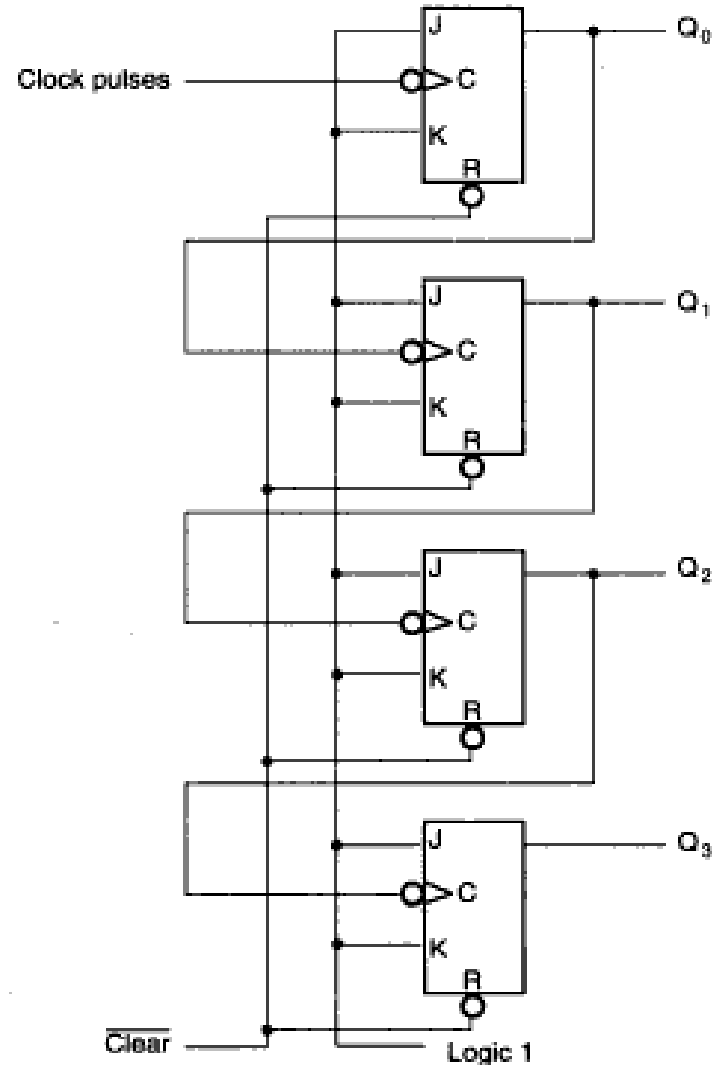
Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

- Now, we need to trigger Q_1 .
- When does Q_1 change value? We need to trigger it then.
- Can we find a pattern between change of value of Q_1 and Q_0 ?
- Q_1 changes value when Q_0 goes from 1 to 0 (-ve change)
- +ve/-ve edge FF?
- Similar for Q_2 and Q_3 .
- What about the last sequence: how it goes from '1111' to '0000'?

Ripple Counter (4-bit Upward Counter using +ve edge D FF)

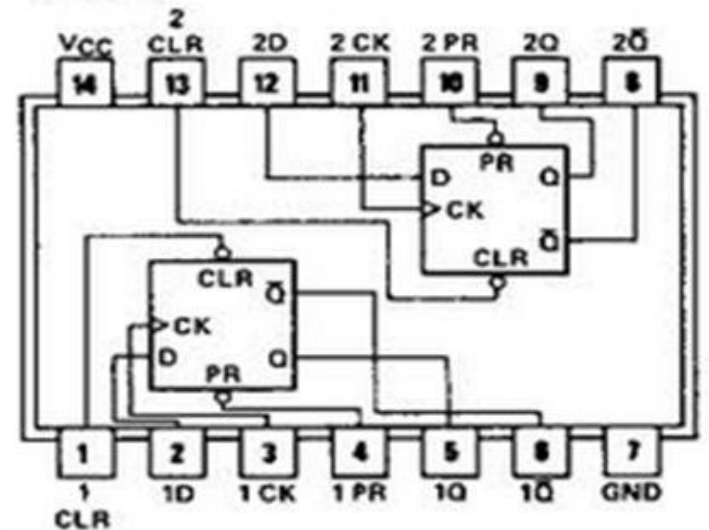
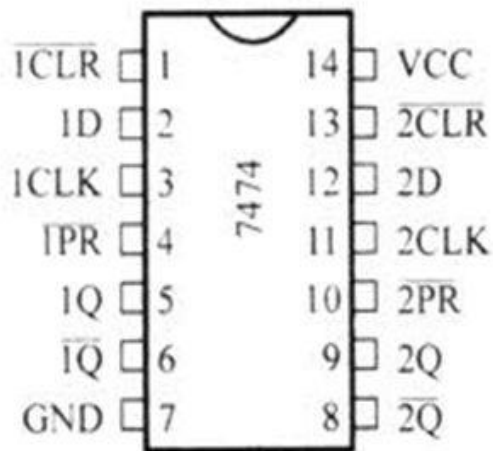


Ripple Counter: 4-bit Upward Counter using -ve edge J-K FF



74AHC74-Dual D Flip-flop

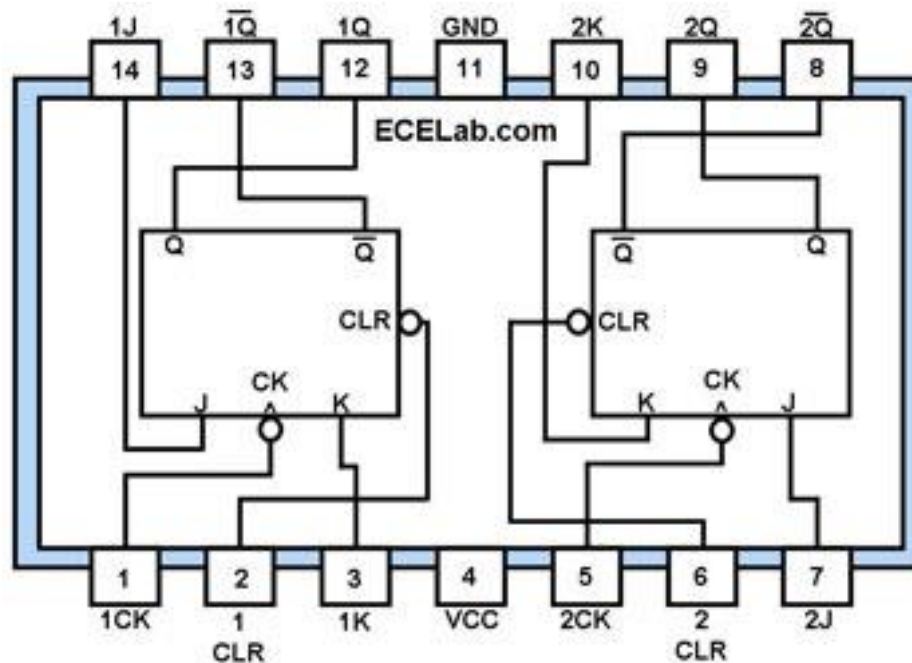
Pin Diagram



Truth Table

ENTREES				SORTIES	
PRESET	CLEAR	CLOCK	D	Q	Q
0	1	X	X	1	0
1	0	X	X	0	1
0	0	X	X	1	1
1	1	↑	1	1	0
1	1	↑	0	0	1
1	1	0	X	Q0	Q0
1	1	1	X	Q0	Q0

7476: Dual J-K Flip-Flops With Preset And Clear



J	K	Next State
		of Q
0	0	Q
0	1	0
1	0	1
1	1	Q-bar

Writing report

- ICs being used
- Pin diagram, Gate diagram of D flip-flop IC
- Logic diagram of 4-bit **downward** counter using
 - +ve edge D flip-flop
 - -ve edge J-K flip-flop